

MODULO 2

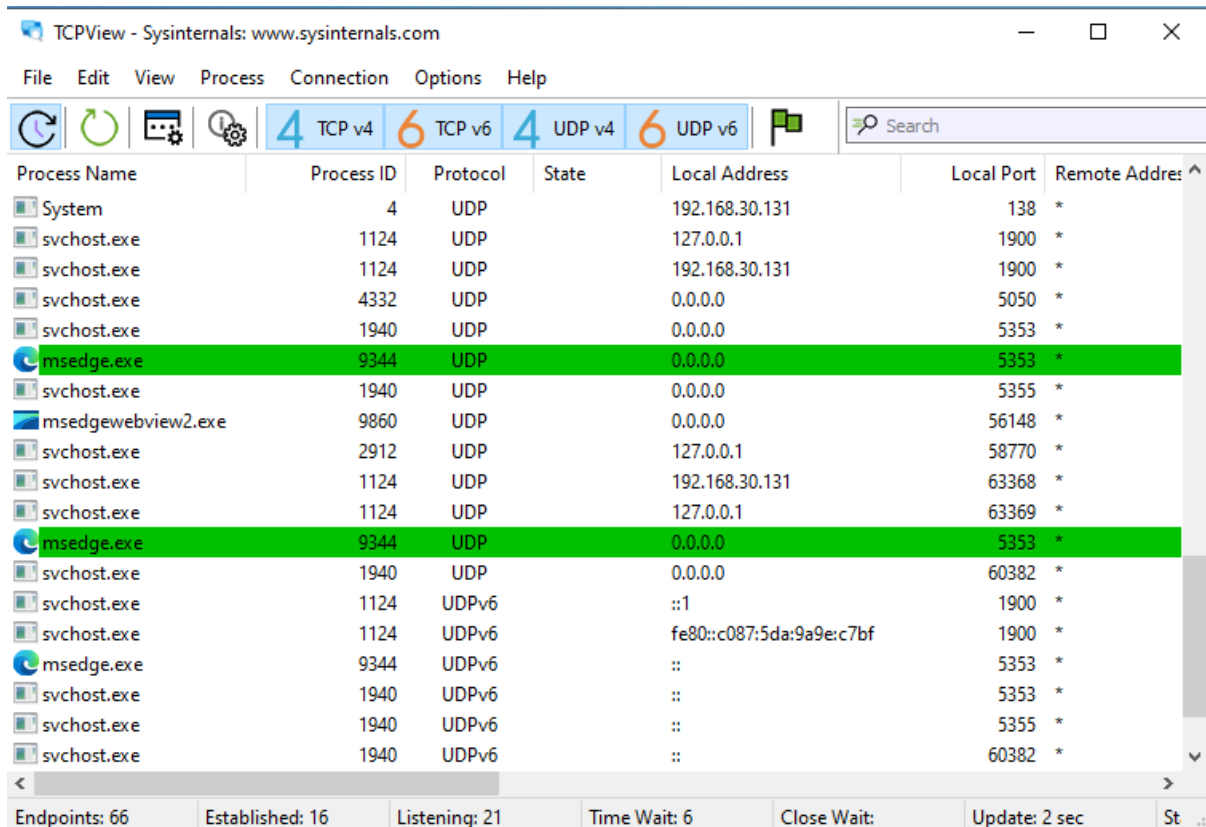
***PRÁCTICA N1 ->**

“Amenazas con Malware”

- Análisis del sistema para detección de malware

- a) El objetivo es utilizar herramientas de monitorización para identificar anomalías que indiquen una infección.

- **TCPView:** Se utilizó para listar todas las conexiones TCP y UDP activas. Permite identificar procesos como msedge.exe comunicándose por red, lo cual es vital para detectar balizas (*beacons*) de un troyano.



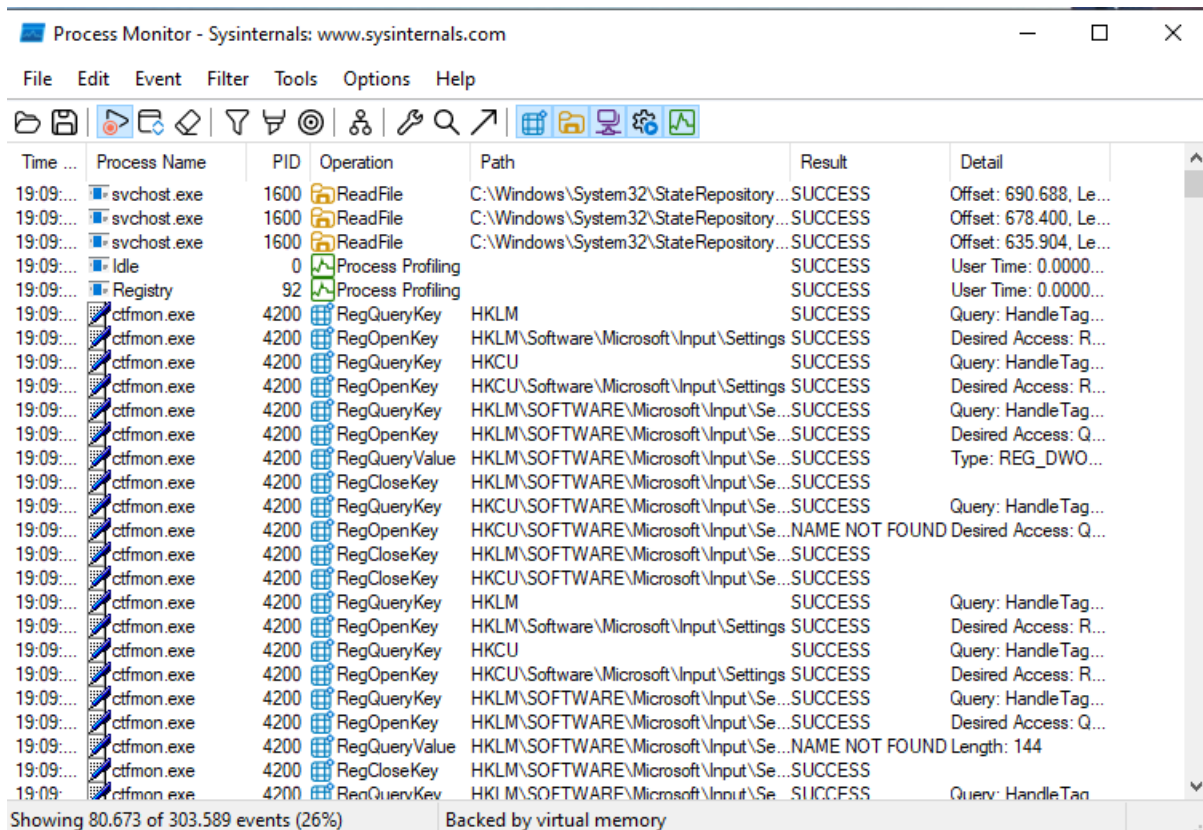
The screenshot shows the TCPView application window. The title bar reads 'TCPView - Sysinternals: www.sysinternals.com'. The menu bar includes 'File', 'Edit', 'View', 'Process', 'Connection', 'Options', and 'Help'. The toolbar contains icons for refreshing, pausing, settings, and a search bar. Below the toolbar, there are filters for '4 TCP v4', '6 TCP v6', '4 UDP v4', and '6 UDP v6'. The main table displays active network connections with columns for Process Name, Process ID, Protocol, State, Local Address, Local Port, and Remote Address. The table is filtered to show only UDP connections. Two connections for msedge.exe are highlighted in green. The status bar at the bottom shows statistics: Endpoints: 66, Established: 16, Listening: 21, Time Wait: 6, Close Wait: 0, Update: 2 sec, and St...

Process Name	Process ID	Protocol	State	Local Address	Local Port	Remote Address
System	4	UDP		192.168.30.131	138	*
svchost.exe	1124	UDP		127.0.0.1	1900	*
svchost.exe	1124	UDP		192.168.30.131	1900	*
svchost.exe	4332	UDP		0.0.0.0	5050	*
svchost.exe	1940	UDP		0.0.0.0	5353	*
msedge.exe	9344	UDP		0.0.0.0	5353	*
svchost.exe	1940	UDP		0.0.0.0	5355	*
msedgewebview2.exe	9860	UDP		0.0.0.0	56148	*
svchost.exe	2912	UDP		127.0.0.1	58770	*
svchost.exe	1124	UDP		192.168.30.131	63368	*
svchost.exe	1124	UDP		127.0.0.1	63369	*
msedge.exe	9344	UDP		0.0.0.0	5353	*
svchost.exe	1940	UDP		0.0.0.0	60382	*
svchost.exe	1124	UDpv6		::1	1900	*
svchost.exe	1124	UDpv6		fe80::c087:5da:9a9e:c7bf	1900	*
msedge.exe	9344	UDpv6		::	5353	*
svchost.exe	1940	UDpv6		::	5353	*
svchost.exe	1940	UDpv6		::	5355	*
svchost.exe	1940	UDpv6		::	60382	*

Endpoints: 66 Established: 16 Listening: 21 Time Wait: 6 Close Wait: 0 Update: 2 sec St...

- **Process Monitor (Procmon):** Se analizó la actividad de los procesos en tiempo

real, observando cómo svchost.exe y ctfmon.exe interactúan con el registro y archivos del sistema.



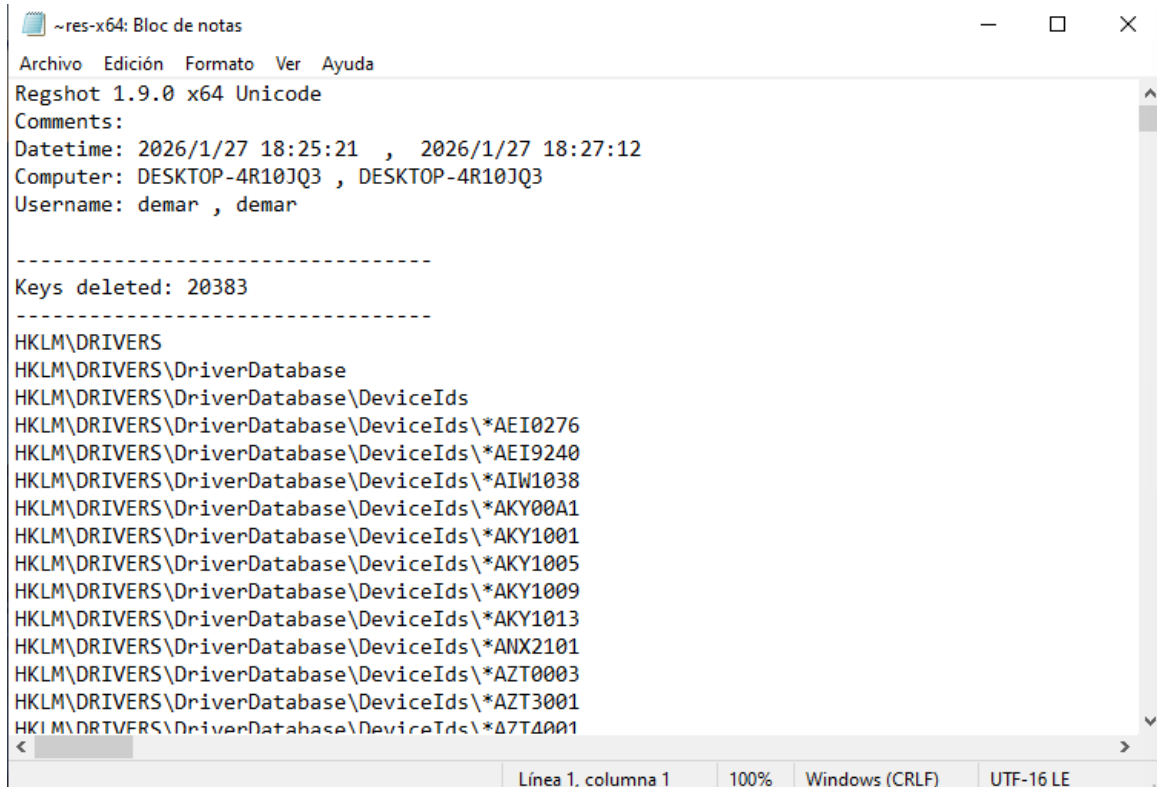
The screenshot shows the Process Monitor application window with a list of events. The events are filtered to show registry operations. The processes involved are svchost.exe and ctfmon.exe. The operations include ReadFile, Process Profiling, RegQueryKey, RegOpenKey, RegCloseKey, and RegQueryValue. The paths are mostly related to the Windows registry, specifically the HKLM\Software\Microsoft\Input\Settings path. The results are mostly SUCCESS, with some NAME NOT FOUND errors. The details column provides additional information about the operations, such as offsets, user time, and query details.

Time ...	Process Name	PID	Operation	Path	Result	Detail
19:09:...	svchost.exe	1600	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 690.688, Le...
19:09:...	svchost.exe	1600	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 678.400, Le...
19:09:...	svchost.exe	1600	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 635.904, Le...
19:09:...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000...
19:09:...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000...
19:09:...	ctfmon.exe	4200	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
19:09:...	ctfmon.exe	4200	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKCU\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
19:09:...	ctfmon.exe	4200	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Desired Access: Q...
19:09:...	ctfmon.exe	4200	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Type: REG_DWO...
19:09:...	ctfmon.exe	4200	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	
19:09:...	ctfmon.exe	4200	RegQueryKey	HKCU\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKCU\SOFTWARE\Microsoft\Input\Se...	NAME NOT FOUND	Desired Access: Q...
19:09:...	ctfmon.exe	4200	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	
19:09:...	ctfmon.exe	4200	RegCloseKey	HKCU\SOFTWARE\Microsoft\Input\Se...	SUCCESS	
19:09:...	ctfmon.exe	4200	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
19:09:...	ctfmon.exe	4200	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKCU\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
19:09:...	ctfmon.exe	4200	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Query: HandleTag...
19:09:...	ctfmon.exe	4200	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Desired Access: Q...
19:09:...	ctfmon.exe	4200	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Se...	NAME NOT FOUND	Length: 144
19:09:...	ctfmon.exe	4200	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	
19:09:...	ctfmon.exe	4200	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Se...	SUCCESS	Query: HandleTag...

Showing 80.673 of 303.589 events (26%) Backed by virtual memory

- **Regshot:** Se realizó un análisis forense del registro comparando dos estados. Se detectó la eliminación masiva de claves (20,383), lo que evidencia una

manipulación profunda del sistema tras una infección.



The screenshot shows a Notepad window titled '~res-x64: Bloc de notas'. The menu bar includes 'Archivo', 'Edición', 'Formato', 'Ver', and 'Ayuda'. The text content is as follows:

```
Regshot 1.9.0 x64 Unicode
Comments:
Datetime: 2026/1/27 18:25:21 , 2026/1/27 18:27:12
Computer: DESKTOP-4R10JQ3 , DESKTOP-4R10JQ3
Username: demar , demar

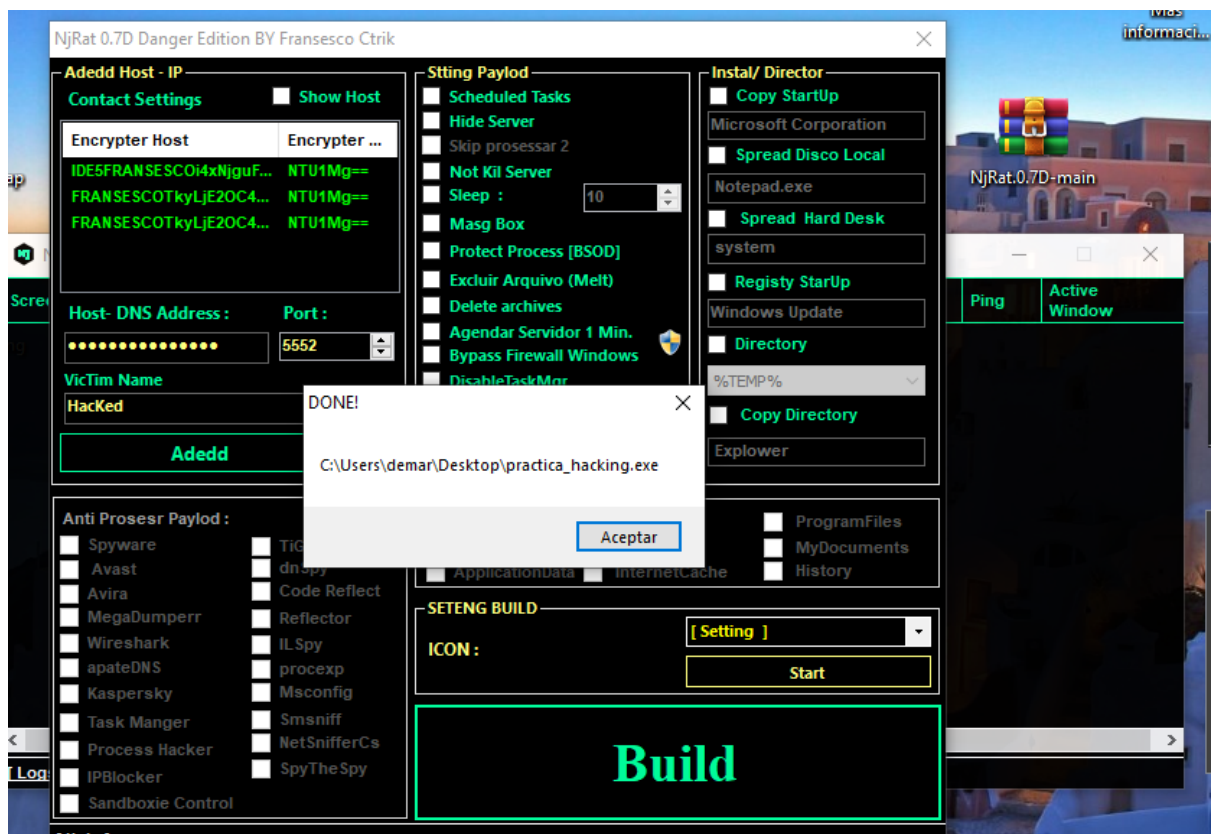
-----
Keys deleted: 20383
-----
HKLM\DRIVERS
HKLM\DRIVERS\DriverDatabase
HKLM\DRIVERS\DriverDatabase\DeviceIds
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AEI0276
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AEI9240
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AIW1038
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AKY00A1
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AKY1001
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AKY1005
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AKY1009
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AKY1013
HKLM\DRIVERS\DriverDatabase\DeviceIds\*ANX2101
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AZT0003
HKLM\DRIVERS\DriverDatabase\DeviceIds\*AZT3001
HKLM\DRIVERS\DriverDatabase\DeviceIds\*A7T4001
```

The status bar at the bottom indicates 'Línea 1, columna 1', '100%', 'Windows (CRLF)', and 'UTF-16 LE'.

B) Control del sistema mediante troyanos (njRAT)

-Se simuló un ataque utilizando un RAT para obtener control total de una máquina víctima.

- **Configuración del Payload:** Se configuró el servidor malicioso practica_hacking.exe con opciones de persistencia y bypass de firewall.



-**Control Remoto:** Tras la ejecución en la víctima, se logró el acceso al escritorio remoto, permitiendo la

monitorización visual de las actividades del usuario.

